

Homework 3 - ADC

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Q.1) Find the SQNR for the following signal quantized uniformly to b bits.

$$A \cos(2\pi f_0 t + \theta_1) + A \sin(2\pi f_0 t + \theta_2)$$

$$x(t) = A \cos(2\pi f_0 t + \theta_1) + A \sin(2\pi f_0 t + \theta_2) \quad \therefore \omega = 2\pi f_0$$

signal Power P_s :

$$\begin{aligned} P_s &= x^2(t) = A^2 (\cos^2(\omega_0 t + \theta_1) + \sin^2(\omega_0 t + \theta_2) + 2\cos(\pi\omega_0 t + \theta_1) \\ &\quad + \sin(\pi\omega_0 t + \theta_2)) \\ &= A^2 \left(\frac{1}{2} + \frac{1}{2} + \sin(\theta_2 - \theta_1) \right) \end{aligned}$$

$$P_s = A^2 (1 + \sin(\theta_2 - \theta_1))$$

Quantization noise power:

$$V = 2R$$

$$\Delta = \frac{V}{2^b} = \frac{2R}{2^b}$$

$$\sigma_q^2 = \frac{\Delta^2}{12} = \frac{(2R/2^b)^2}{12} = \frac{R^2}{3(2^b)}$$

SQNR:

$$SQNR = \frac{P_{av}}{\sigma_q^2} = \frac{R^2/2}{R^2/3(2^b)} = \frac{3}{2} \cdot 2^{2b}$$

$$SQNR_{dB} = 10 \log_{10} \left(\frac{3}{2} \cdot 2^{2b} \right) = \boxed{1.76 + 6.02b \text{ dB}}$$

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Q.2) Find the numerical value of SQNR if the parameters are:
 $A = 1 \text{ V}$, $f_0 = 60 \text{ Hz}$, $\theta_1 = \frac{\pi}{2} \text{ rads}$, $\theta_2 = \frac{3\pi}{2} \text{ rads}$ and $b = 8 \text{ bits}$.

$$\rightarrow \text{Phase difference: } \frac{3\pi}{2} - \frac{\pi}{2} = \pi$$

$$\rightarrow \sin(\pi) = 0$$

Signal Power:

$$\rightarrow P_s = A^2 (1 + \sin(\theta_2 - \theta_1)) = 1^2 (1 + 0) = 1$$

$$\rightarrow \text{SQNR: } \frac{3}{2} \cdot 2^{2b} = \frac{3}{2} \cdot 2^{2(8)} = 98304$$

$$\rightarrow \text{SQNR}_{\text{dB}}: 10 \log_{10}(98304) = 49.9 \text{ dB}$$

Q.3) Repeat parts (1) and (2) for a Gaussian distributed random signals with $\sigma = 1$ and the quantizer limited to $[-3/2 \text{ V}, 3/2 \text{ V}]$.

$$\rightarrow \text{Signal Power: } P_s = \sigma^2 = 1^2 = 1$$

$$\rightarrow \text{Step size: Range} = 3 \text{ V}, \text{ levels} = 2^8 = 256$$

$$\Delta = \frac{V}{2^b} = \frac{3}{2^8} = \frac{3}{256} = 0.01172 \text{ V}$$

$$\rightarrow \text{Quantization noise power: } \sigma_q^2 = \frac{\Delta^2}{12} = \frac{(0.01172)^2}{12} = 1.14 \times 10^{-5}$$

$$\rightarrow \text{SQNR (Linear): } \frac{P_s}{\sigma_q^2} = \frac{1}{1.14 \times 10^{-5}} = 87407$$

$$\rightarrow \text{SQNR (dB): } 10 \log_{10}(87407) = 49.41 \text{ dB}$$

Signal Analysis

Q.1) Prove that if a signal $x(t)$ is real-valued and has a conjugate symmetric spectrum ($X(f) = X^*(-f)$) then its Fourier magnitude is an even function or,

$$|X(f)| = |X(-f)|$$

without using integrals.

$$\rightarrow X(f) = X^*(-f)$$

$$\rightarrow X^*(f) = X(-f)$$

$$\rightarrow |X(f)|^2 = X(f) \cdot X^*(f) \\ = X(f) X(-f)$$

$$\rightarrow |X(-f)|^2 = X(-f) \cdot X^*(-f) \\ = X(-f) \cdot X(f)$$

$$\rightarrow |X(f)|^2 = |X(-f)|^2$$

$$\Rightarrow |X(f)| = |X(-f)|$$

Q.2) Prove that for above such signals the phase is an odd function or anti-symmetric, i.e.:

$$\arg X(f) = -\arg X(-f)$$

$$\rightarrow X(f) = |X(f)| e^{j\phi(f)}$$

$$\rightarrow X(-f) = |X(-f)| e^{j\phi(-f)}$$

$$\rightarrow X(f) = X^*(-f)$$

$$\hookrightarrow |X(f)| e^{j\phi(f)} = (|X(-f)| e^{j\phi(-f)})^*$$

$$\rightarrow (|X(-f)| e^{j\phi(-f)})^* = |X(-f)| e^{-j\phi(-f)}$$

$$\rightarrow |X(f)| e^{j\phi(f)} = |X(-f)| e^{-j\phi(-f)}$$

$$\Rightarrow |X(f)| = |X(-f)|$$

$$e^{j\phi(f)} = e^{-j\phi(-f)} \Rightarrow \phi(f) = -\phi(-f)$$

$$\Rightarrow \arg X(f) = -\arg X(-f)$$

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Q.3) Prove the converse: If the FT is conjugate symmetric then the signal is real valued.

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} \cdot df$$

$$x^*(t) = \left(\int_{-\infty}^{\infty} X(f) e^{j2\pi ft} \cdot df \right)^*$$

$$\therefore (ab^*) = a^* b^*$$

$$\therefore (e^{j2\pi ft})^* = e^{-j2\pi ft}$$

$$x^*(t) = \int_{-\infty}^{\infty} X^*(f) e^{-j2\pi ft} \cdot df$$

$$\therefore X^*(f) = X(-f)$$

$$\rightarrow x^*(t) = \int_{-\infty}^{\infty} X(-f) e^{-j2\pi ft} \cdot df$$

$$u = -f, \quad df = -du$$

$$\rightarrow x^*(t) = \int_{-\infty}^{\infty} X(u) e^{j2\pi ut} \cdot du$$

$$x^*(t) = x(t)$$

Q.4) Verify the following relationship between the spectra of the bandpass and the corresponding baseband signal:

$$X(f) = \frac{1}{2} \{ X_{bb}(f - f_c) + X_{bb}^*(-f - f_c) \}$$

$$x(t) = \frac{1}{2} x_{bb}(t) e^{j2\pi f_c t} + \frac{1}{2} x_{bb}^*(t) e^{-j2\pi f_c t}$$

$$X(f) = \frac{1}{2} \mathcal{F} \{ x_{bb}(t) e^{j2\pi f_c t} \} + \frac{1}{2} \mathcal{F} \{ x_{bb}^*(t) e^{-j2\pi f_c t} \}$$

$$\begin{aligned}
 & \bullet \quad x(t) e^{j2\pi f_c t} \xrightarrow{f} X(f - f_c) \\
 & \quad \mathcal{F} \{ x_{bb}(t) e^{j2\pi f_c t} \} = X_{bb}(f - f_c) \\
 & \bullet \quad x_{bb}^*(t) \xrightarrow{f} X_{bb}^*(-f) \\
 & \quad \mathcal{F} \{ x_{bb}^*(t) e^{-j2\pi f_c t} \} = X_{bb}^*(-f - f_c)
 \end{aligned}$$

$$X(f) = \frac{1}{2} [X_{bb}(f - f_c) + X_{bb}^*(-f - f_c)]$$

Q.5) Establish the following identity using what was taught in the lectures:

$$\begin{aligned}
 X(t) &= \text{Re} \{ x_I(t) + j x_Q(t) \} [\cos(2\pi f_c t) + j \sin(2\pi f_c t)] \\
 &= \text{Re} \{ x_{bb}(t) e^{j2\pi f_c t} \}
 \end{aligned}$$

$$x_{bb}(t) = x_I(t) + j x_Q(t)$$

$$\begin{aligned}
 x_{bb}(t) e^{j2\pi f_c t} &= (x_I + j x_Q) [\cos(2\pi f_c t) + j \sin(2\pi f_c t)] \\
 &= x_I \cos(2\pi f_c t) + j x_I \sin(2\pi f_c t) + j x_Q \cos(2\pi f_c t) - x_Q \sin(2\pi f_c t) \\
 &= \underbrace{(x_I \cos(2\pi f_c t) - x_Q \sin(2\pi f_c t))}_{\text{real}} + j \underbrace{(x_Q \cos(2\pi f_c t) + x_I \sin(2\pi f_c t))}_{\text{imag.}}
 \end{aligned}$$

$$\text{Re} \{ x_{bb}(t) e^{j2\pi f_c t} \} = \text{Re} \{ x_I \cos(2\pi f_c t) - x_Q \sin(2\pi f_c t) \}$$

Q.) Inner Product and Orthogonality

Prove that $x_I(t) \cos 2\pi f_c t \perp x_Q(t) \sin 2\pi f_c t$ using Parseval's.

$$\begin{aligned}
 \text{Parseval's Theorem: } \langle x(t), y(t) \rangle &= \int_{-\infty}^{\infty} x(t) y^*(t) \cdot dt \\
 &= \int_{-\infty}^{\infty} X(f) Y^*(f) \cdot df
 \end{aligned}$$

$$\langle x_I \cos(2\pi f_c t), x_Q \sin(2\pi f_c t) \rangle = \int_{-\infty}^{\infty} X_I(f) [X_Q(f)]^* df$$

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$$\text{since, } \cos(2\pi f_c t) \xrightarrow{f} \frac{1}{2} [\delta(f-f_c) + \delta(f+f_c)]$$

$$\rightarrow X_I(f) = \frac{1}{2} [X_I(f-f_c) + X_I(f+f_c)]$$

$$\text{since, } \sin(2\pi f_c t) \xrightarrow{f} \frac{1}{2j} [\delta(f-f_c) - \delta(f+f_c)]$$

$$\rightarrow X_Q(f) = \frac{1}{2j} [X_Q(f-f_c) - X_Q(f+f_c)]$$

$$X_I(f) [X_Q(f)]^* = \frac{1}{4j} [X_I(f-f_c) + X_I(f+f_c)] [X_Q^*(f-f_c) - X_Q^*(f+f_c)]$$

$$= \frac{1}{4j} [X_I(f-f_c) X_Q^*(f-f_c) - \cancel{X_I(f-f_c) X_Q^*(f+f_c)} + \cancel{X_I(f+f_c) X_Q^*(f-f_c)} - X_I(f+f_c) X_Q^*(f+f_c)]$$

$$= \frac{1}{4j} [X_I(f-f_c) X_Q^*(f-f_c) - X_I(f+f_c) X_Q^*(f+f_c)]$$

$$\therefore u = f-f_c, \quad v = f+f_c$$

$$\int X_I(u) X_Q^*(u) \cdot du - \int X_I(v) X_Q^*(v) \cdot dv = 0$$

$$\langle X_I(t) \cos(2\pi f_c t), X_Q(t) \sin(2\pi f_c t) \rangle = 0$$

(p.) One-sided signals

Prove that $x_+(t) + x_-(t) = x(t)$ and hence establish that $x_+(t)$ and $x_-(t)$ are conjugate pairs.

$$X_+(f) = \begin{cases} X(f) & , f > 0 \\ 0 & , f < 0 \end{cases}$$

$$X_-(f) = \begin{cases} 0 & , f > 0 \\ X(f) & , f < 0 \end{cases}$$

$$X(f) = X_+(f) + X_-(f)$$

Taking the inverse fourier transform:

$$x_+(t) = f^{-1} \{X_+(f)\}, \quad x_-(t) = f^{-1} \{X_-(f)\}$$

$$x_+(t) + x_-(t) = f^{-1} \{X_+(f)\} + f^{-1} \{X_-(f)\}$$

$$= f^{-1} \{X(f)\} = x(t)$$

$$X(f) = X^*(-f)$$

$f > 0$:

$$X_+(f) = X(f), \quad X_-(f) = 0$$

$f < 0$:

$$X_-(f) = X(f) = X^*(-f) = X_+^*(-f)$$

$$X_-(f) = X_+^*(-f)$$

taking fourier transform:

$$x_-(t) = x_+^*(t)$$

Q.) Hilbert Transform(er)

Establish that the transfer function of Hilbert transformer:

$$H(f) = -j \operatorname{sign}(f)$$

and also, that the impulse response is given by:

$$h(t) = \frac{1}{\pi t} \quad -\infty < t < \infty$$

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- What is single fact that makes this filter un-realizable in the analog domain?
- What are the two facts that together that make this filter un-realizable in the digital domain as well?
- What approximation could we make to implement the Hilbert Transformer in digital domain? In the analog signal?

From chapter 3 eq. 3.38 - 3.39, the Hilbert Transform satisfies:

$$\mathcal{F}\{x(t)\} = -j \operatorname{sgn}(f) X(f)$$

$\therefore$ the freq response of the hilbert transform is;

$$H(f) = -j \operatorname{sgn}(f)$$

$$\operatorname{sgn}(f) = \begin{cases} 1 & , f > 0 \\ 0 & , f = 0 \\ -1 & , f < 0 \end{cases}$$

$$h(t) = \mathcal{F}^{-1}\{-j \operatorname{sgn}(f)\}$$

from book eq. (3.46): $h(t) = \frac{1}{\pi t}$

- The single fact that makes this filter un-realizable in analog domain is the impulse response $h(t) = \frac{1}{\pi t}$ is non-causal.
- It is unrealizable in the digital domain bcs its non-causal and has infinite duration.
- In digital domain: Implement an approximate Hilbert transform by the impulse response to a finite length and applying a time delay to the system.

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→ In analog domain: We can all-pass 90° phase shifter constructed from: RC-CR network; Polyphase all-pass network; Balanced RC lattice filter.